Audio Noise Reduction Using FIR Window Techniques

Ved Dhandhukia
TYETC14, Group 13
Dept. of Electronics and Telecommunication Engg.
Shah & Anchor Kutchhi Engineering College
Mumbai, India

Aditya Joshi
TYETC30, Group 13
Dept. of Electronics and Telecommunication Engg.
Shah & Anchor Kutchhi Engineering College
Mumbai, India

***Abstract*—Audio recordings are often corrupted by broadband high-frequency noise such as microphone hiss and electronic interference, which masks tonal content and reduces quality. This paper designs three linear-phase FIR low-pass filters using the Hamming, Hann and Blackman windows and compares them on a synthetic five-tone signal sampled at 16 kHz with added band-limited noise. The filters (order 100, cutoff 3.4 kHz) are evaluated using time-domain plots, FFT spectra, frequency responses and SNR improvement. All three filters preserve the five tones and suppress the noise band; the Blackman window gives the deepest stopband and the highest measured SNR improvement (17.02 dB), followed by Hann (16.86 dB) and Hamming (16.82 dB), at the cost of a wider transition band.**

***Index Terms*—** *FIR filter, window method, Hamming, Hann, Blackman, noise reduction, FFT, SNR improvement, linear phase.*

I. Introduction

Noise suppression is a front-end step in music analysis, speech communication, hearing-assistive devices and pitch detection, where noise degrades downstream accuracy. FIR filters are attractive for this task because they are always stable and, with symmetric coefficients, have exactly linear phase, so the waveform shape of the useful signal is not distorted. In the window method, the choice of window sets the trade-off between transition-band sharpness, stopband attenuation and filter length, which makes it a practical design decision.

The objective of this work is to suppress high-frequency noise in an audio signal sampled at 16 kHz without distorting its tonal components, and to rank the Hamming, Hann and Blackman windows by stopband attenuation, transition width and SNR improvement. Hardware/DSP-processor implementation, adaptive or IIR filtering, real-time operation and fixed-point effects are outside the scope of this work.

II. Literature Review

Oppenheim and Schafer [1] present the window method, in which an ideal low-pass impulse response is truncated and multiplied by a window, and give the standard trade-offs: minimum stopband attenuation of about 44 dB for Hann, 53 dB for Hamming and 74 dB for Blackman, with a wider transition band as attenuation increases. Harris [2] compares windows by main-lobe width, peak sidelobe level and sidelobe roll-off, and explains spectral leakage and why Blackman suppresses leakage better at the cost of resolution. Proakis and Manolakis [3] cover linear-phase FIR types and the relation between filter length and transition width. Mitra [4] provides MATLAB-style design workflows (e.g., `fir1`) and filter-structure details.

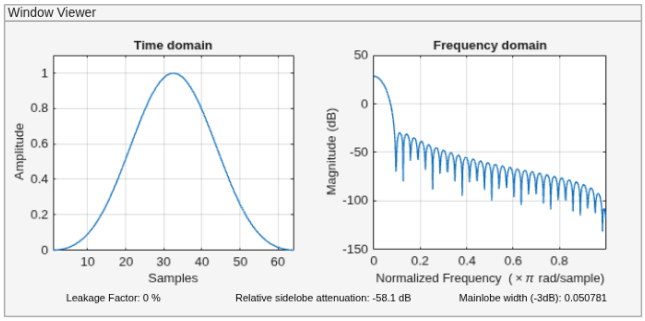


Fig. 1. Time-domain and frequency-domain views of the Blackman window (MATLAB Window Viewer).

These references treat windows theoretically. This work verifies the trade-offs on a noisy multi-tone signal, using the measured SNR improvement as a single performance metric.

III. Theory and Design Methodology

A. Window-Based FIR Design

The ideal low-pass impulse response with cutoff ω_c and delay $M = (N-1)/2$ is

$$h_d[n] = \frac{\sin\big(\omega_c(n-M)\big)}{\pi(n-M)}, \tag{1}$$

which is truncated to length N and multiplied by a window $w[n]$:

$$h[n] = h_d[n]\,w[n], \qquad 0 \le n \le N-1. \tag{2}$$

Symmetry of $h[n]$ gives a Type I linear-phase filter with a constant group delay of $(N-1)/2$ samples. A window with larger sidelobe suppression has a wider main lobe, which results in a wider transition band. For example, the Hann window is $w[n] = 0.5 - 0.5\cos\big(2\pi n/(N-1)\big)$. The time- and frequency-domain views of the Blackman window are shown in Fig. 1.

B. Performance Measure

With the clean signal $x[n]$ as reference, the SNR in dB is

$$\mathrm{SNR} = 10\log_{10}\frac{\sum x^2[n]}{\sum (y[n]-x[n])^2}, \tag{3}$$

and the improvement is $\mathrm{SNR}_{\mathrm{out}} - \mathrm{SNR}_{\mathrm{in}}$. The output is advanced by the filter delay ($d = 50$ samples) before comparison with the clean reference.

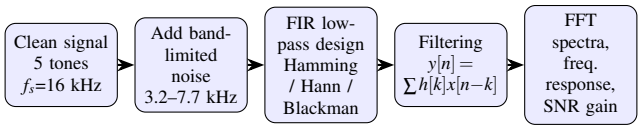


Fig. 2. Block diagram of the proposed system.

C. Test Signal and Noise

The clean signal is the sum of five sinusoids at 180, 450, 900, 1800 and 2800 Hz, normalised to unit peak amplitude, with $f_s = 16$ kHz (Nyquist 8 kHz) and a duration of 1 s. The noise is white Gaussian noise passed through a 6th-order Butterworth band-pass filter (3.2–7.7 kHz, applied with zero-phase `filtfilt`), so its energy lies above the signal band. It is scaled relative to the signal RMS before being added.

D. Filter Specification

Linear-phase FIR low-pass filters of order 100 (101 taps) with a cutoff of 3.4 kHz are designed with the Hamming, Hann and Blackman windows at equal length. The passband covers all five tones (up to 2.8 kHz) and the stopband extends to 8 kHz. The workflow is shown in Fig. 2.

E. Implementation

The system was implemented in MATLAB. The key design and filtering steps are listed below.

```
fs = 16000; t = 0:1/fs:1-1/fs;
x = sin(2*pi*180*t) + sin(2*pi*450*t) ...
   + sin(2*pi*900*t) + sin(2*pi*1800*t) ...
   + sin(2*pi*2800*t);
x = x/max(abs(x));
rng(42); n = randn(size(t));
[b,a] = butter(6,[3200 7700]/(fs/2),'bandpass');
n = filtfilt(b,a,n); n = n/rms(n);
n = n*rms(x)*sqrt(10^(5/10));
noisy = x + n;
N = 100; fc = 3400;
b1 = fir1(N,fc/(fs/2),hamming(N+1));
b2 = fir1(N,fc/(fs/2),hann(N+1));
b3 = fir1(N,fc/(fs/2),blackman(N+1));
y1 = filter(b1,1,noisy);
y2 = filter(b2,1,noisy);
y3 = filter(b3,1,noisy);
```

IV. Results and Discussion

A. Time-Domain Comparison

Fig. 3 shows the clean signal, the noisy input and the Blackman-filtered output over 0.04 s. The filtered output follows the clean waveform closely and is much smoother than the noisy input. The flat segment at the start of the filtered output is the start-up transient of the 50-sample filter delay.

B. Frequency Spectrum

In Fig. 4 the five spectral peaks of the clean signal are preserved in the filtered output, while the noise band above about 3 kHz is strongly attenuated.

C. Filter Frequency Responses

Fig. 5 compares the three filters. All pass the signal up to about 3 kHz. The Hamming filter has a constant, shallower stopband ripple of roughly -55 to -64 dB, whereas the Hann and Blackman filters fall to much lower levels (about -100 dB and below at high frequencies). The Blackman filter has the deepest stopband and the widest transition band.

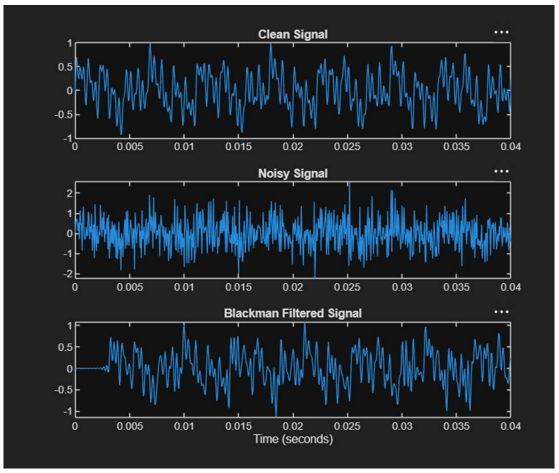


Fig. 3. Clean, noisy and Blackman-filtered signals (time domain).

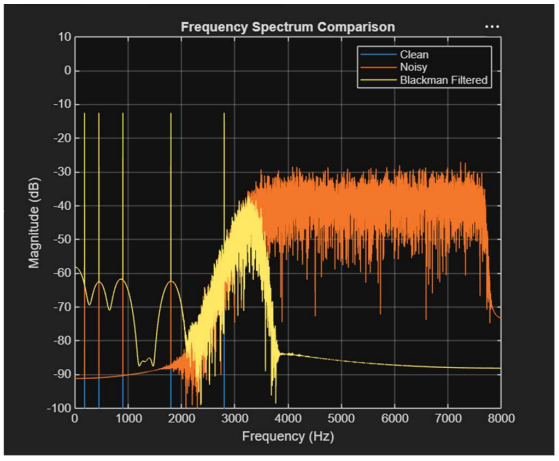


Fig. 4. Frequency spectrum of the clean, noisy and Blackman-filtered signals.

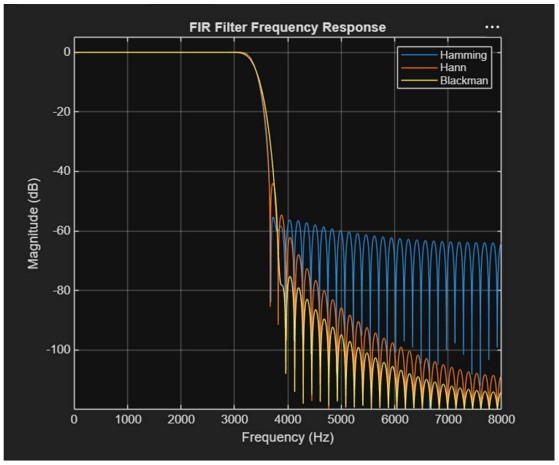


Fig. 5. Frequency responses of the Hamming, Hann and Blackman FIR filters.

TABLE I
WINDOW COMPARISON (ORDER 100, f_c = 3.4 KHZ)

Window	Stopband	Transition	SNR gain (dB)
Hamming	Lowest (55–64 dB)	Narrowest	16.82
Hann	Intermediate	Intermediate	16.86
Blackman	Highest	Widest	17.02

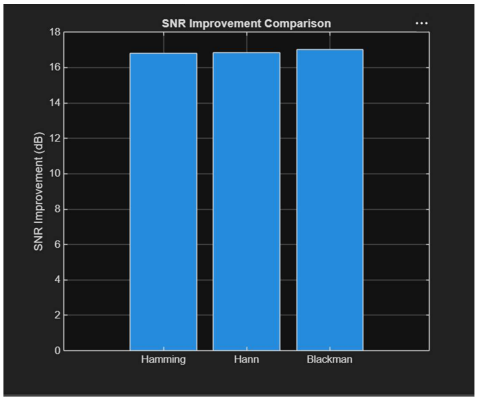


Fig. 6. SNR improvement for the three window techniques.

D. SNR Improvement

The measured SNR improvements are given in Table I and Fig. 6. Blackman achieves the highest improvement, but the differences between the windows are small (< 0.3 dB), because the tones lie well below the cutoff and most of the noise lies above it. The window choice therefore depends on the application: Blackman is preferred when stopband cleanup matters most, and Hamming when a sharper transition is preferred.

V. LIMITATIONS

1. The experiment uses a synthetic signal rather than real audio.
2. The filter targets high-frequency noise and cannot remove low-frequency or overlapping noise.
3. Useful audio components above the cutoff frequency are also attenuated.
4. FIR filtering introduces a delay of $(N-1)/2 = 50$ samples, which may affect real-time applications.
5. Performance depends on the cutoff, filter order, window and noise characteristics.

VI. CONCLUSION

Window-based FIR low-pass filters reduce high-frequency noise while retaining the five designed tones. Among the three windows, Blackman gives the deepest stopband and the highest SNR improvement (17.02 dB) for the tested synthetic signal, at the cost of the widest transition band, while Hamming offers the sharpest transition. Future work includes validation on real audio recordings, other noise types, and a study of filter-order and delay trade-offs (e.g., Kaiser window and Parks–McClellan designs).

ACKNOWLEDGMENT

The authors thank the Department of Electronics and Telecommunication Engineering, Shah & Anchor Kutchhi Engineering College, for support of this DTSP micro-project.

REFERENCES

[1] A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*, 3rd ed. Pearson, 2010.

[2] F. J. Harris, "On the use of windows for harmonic analysis with the discrete Fourier transform," *Proc. IEEE*, vol. 66, no. 1, pp. 51–83, Jan. 1978.

[3] J. G. Proakis and D. G. Manolakis, *Digital Signal Processing: Principles, Algorithms and Applications*, 4th ed. Pearson, 2007.

[4] S. K. Mitra, *Digital Signal Processing: A Computer-Based Approach*. McGraw-Hill.